

DSA Rehabilitation Plan

132 problems: 103 from Codeforces and 29 from LeetCode (only the classics that matter). Ratings are approximate. Codeforces IDs are contest + letter (e.g. 1352C = contest 1352, problem C). Work top to bottom within each section; difficulty rises.

1. Implementation / warm-up

Done	ID	Problem	Source	Rating
☐	1A	Theatre Square	Codeforces	1000
☐	4A	Watermelon	Codeforces	800
☐	50A	Domino Piling	Codeforces	800
☐	263A	Beautiful Matrix	Codeforces	800
☐	112A	Petya and Strings	Codeforces	800
☐	118A	String Task	Codeforces	1000
☐	1328A	Divisibility Problem	Codeforces	800
☐	1409A	Yet Another Two Integers Problem	Codeforces	800
☐	1475A	Odd Divisor	Codeforces	900
☐	1352C	K-th Not Divisible by n	Codeforces	1200

2. Sorting and greedy

Done	ID	Problem	Source	Rating
☐	160A	Twins	Codeforces	900
☐	34B	Sale	Codeforces	900
☐	1472D	Even-Odd Game	Codeforces	1200
☐	1360B	Honest Coach	Codeforces	800
☐	977C	Less or Equal	Codeforces	1200
☐	1409B	Minimum Product	Codeforces	1100
☐	1374B	Multiply by 2, divide by 6	Codeforces	900
☐	1335C	Two Teams Composing	Codeforces	1200
☐	1196B	Odd Sum Division	Codeforces	1200
☐	1294C	Product of Three Numbers	Codeforces	1300
☐	1352B	Same Parity Summands	Codeforces	1200
☐	1359B	New Theatre Square	Codeforces	1000
☐	1486B	Eastern Exhibition	Codeforces	1500
☐	1348C	Phoenix and Distribution	Codeforces	1400
☐	1409D	Decrease the Sum of Digits	Codeforces	1500

3. Math, number theory, combinatorics

Done	ID	Problem	Source	Rating
☐	26A	Almost Prime	Codeforces	900
☐	230B	T-primes	Codeforces	1300

Done	ID	Problem	Source	Rating
☐	1474B	Different Divisors	Codeforces	1000
☐	1475B	New Year's Number	Codeforces	900
☐	489C	Given Length and Sum of Digits	Codeforces	1400
☐	1370C	Number Game	Codeforces	1400
☐	1285C	Fadi and LCM	Codeforces	1400
☐	1342E	Placing Rooks	Codeforces	2200

4. Prefix sums, hashing, maps

Done	ID	Problem	Source	Rating
☐	313B	Ilya and Queries	Codeforces	1100
☐	816B	Karen and Coffee	Codeforces	1400
☐	18C	Stripe	Codeforces	1400
☐	4C	Registration System	Codeforces	1300
☐	1141C	Polycarp Restores Permutation	Codeforces	1200
☐	1520D	Same Differences	Codeforces	1200
☐	474B	Worms	Codeforces	1200
☐	466C	Number of Ways	Codeforces	1700
☐	560	Subarray Sum Equals K	LeetCode	-

5. Binary search, two pointers, sliding window

Done	ID	Problem	Source	Rating
☐	706B	Interesting Drink	Codeforces	1100
☐	279B	Books	Codeforces	1400
☐	1352D	Alice, Bob and Candies	Codeforces	1300
☐	1538C	Number of Pairs	Codeforces	1300
☐	702C	Cellular Network	Codeforces	1500
☐	1201C	Maximum Median	Codeforces	1400
☐	371C	Hamburgers	Codeforces	1600
☐	1073C	Vasya and Robot	Codeforces	1800
☐	3	Longest Substring Without Repeating Characters	LeetCode	-
☐	76	Minimum Window Substring	LeetCode	-
☐	33	Search in Rotated Sorted Array	LeetCode	-
☐	875	Koko Eating Bananas	LeetCode	-

6. Stack, queue, monotonic structures, heaps

Done	ID	Problem	Source	Rating
☐	5C	Longest Regular Bracket Sequence	Codeforces	2000
☐	1374C	Move Brackets	Codeforces	1000
☐	547B	Mike and Feet	Codeforces	1900
☐	1353D	Constructing the Array	Codeforces	1700

Done	ID	Problem	Source	Rating
☐	20	Valid Parentheses	LeetCode	-
☐	739	Daily Temperatures	LeetCode	-
☐	84	Largest Rectangle in Histogram	LeetCode	-
☐	239	Sliding Window Maximum	LeetCode	-
☐	215	Kth Largest Element	LeetCode	-
☐	295	Find Median from Data Stream	LeetCode	-

7. Linked list and design

Done	ID	Problem	Source	Rating
☐	206	Reverse Linked List	LeetCode	-
☐	141	Linked List Cycle	LeetCode	-
☐	146	LRU Cache	LeetCode	-
☐	208	Implement Trie	LeetCode	-

8. Strings (KMP/Z, trie, XOR trie)

Done	ID	Problem	Source	Rating
☐	550A	Two Substrings	Codeforces	1500
☐	126B	Password	Codeforces	1700
☐	706D	Vasiliy's Multiquery	Codeforces	1800
☐	49	Group Anagrams	LeetCode	-

9. Segment tree and Fenwick tree

Done	ID	Problem	Source	Rating
☐	339D	Xenia and Bit Operations	Codeforces	1700
☐	459D	Pashmak and Parmida's Problem	Codeforces	1800
☐	61E	Enemy is Weak	Codeforces	1900
☐	380C	Sereja and Brackets	Codeforces	2000
☐	52C	Circular RMQ	Codeforces	2200
☐	1354D	Multiset	Codeforces	1900

10. Dynamic programming

Classic DP

Done	ID	Problem	Source	Rating
☐	189A	Cut Ribbon	Codeforces	1300
☐	455A	Boredom	Codeforces	1500
☐	1472C	Long Jumps	Codeforces	1100
☐	1105C	Ayoub and Lost Array	Codeforces	1500
☐	166E	Tetrahedron	Codeforces	1500
☐	706C	Hard Problem	Codeforces	1600

Done	ID	Problem	Source	Rating
☐	1389B	Array Walk	Codeforces	1600
☐	322	Coin Change	LeetCode	-
☐	300	Longest Increasing Subsequence	LeetCode	-
☐	1143	Longest Common Subsequence	LeetCode	-
☐	72	Edit Distance	LeetCode	-
☐	416	Partition Equal Subset Sum	LeetCode	-
☐	139	Word Break	LeetCode	-

Bitmask / SOS DP

Done	ID	Problem	Source	Rating
☐	580D	Kefa and Dishes	Codeforces	1800
☐	8C	Looking for Order	Codeforces	2000
☐	453B	Little Pony and Harmony Chest	Codeforces	2000
☐	165E	Compatible Numbers	Codeforces	2200

11. Graphs

BFS / DFS

Done	ID	Problem	Source	Rating
☐	520B	Two Buttons	Codeforces	1400
☐	115A	Party	Codeforces	900
☐	217A	Ice Skating	Codeforces	1200
☐	862B	Mahmoud and Ehab and the Bipartiteness	Codeforces	1500
☐	1176E	Cover it!	Codeforces	1700
☐	200	Number of Islands	LeetCode	-

DSU

Done	ID	Problem	Source	Rating
☐	277A	Learning Languages	Codeforces	1400
☐	977E	Cyclic Components	Codeforces	1500
☐	1139C	Edgy Trees	Codeforces	1500
☐	1213G	Path Queries	Codeforces	1900
☐	25D	Roads not only in Berland	Codeforces	2200

Shortest paths

Done	ID	Problem	Source	Rating
☐	25C	Roads in Berland	Codeforces	1500
☐	20C	Dijkstra?	Codeforces	1900
☐	1650G	Counting Shortcuts	Codeforces	1900
☐	59E	Shortest Path	Codeforces	2000
☐	545E	Paths and Trees	Codeforces	2000
☐	1472G	Moving to the Capital	Codeforces	2000

MST

Done	ID	Problem	Source	Rating
☐	1245D	Shichikuji and Power Grid	Codeforces	1900
☐	160D	Edges in MST	Codeforces	2200

Topological sort

Done	ID	Problem	Source	Rating
☐	510C	Fox and Names	Codeforces	1600
☐	1385E	Directing Edges	Codeforces	2000
☐	207	Course Schedule	LeetCode	-

12. Trees

Done	ID	Problem	Source	Rating
☐	580C	Kefa and Park	Codeforces	1500
☐	161D	Distance in Tree	Codeforces	1800
☐	1092F	Tree with Maximum Cost	Codeforces	1800
☐	1101D	GCD Counting	Codeforces	1900
☐	1328E	Tree Queries	Codeforces	1900
☐	1304E	1-Trees and Queries	Codeforces	2000
☐	337D	Book of Evil	Codeforces	2000
☐	600E	Lomsat gelral	Codeforces	2300
☐	236	Lowest Common Ancestor of a Binary Tree	LeetCode	-

13. Bit manipulation, backtracking, meet-in-the-middle

Done	ID	Problem	Source	Rating
☐	276D	Little Girl and Maximum XOR	Codeforces	1900
☐	1006F	Xor-Paths	Codeforces	2000
☐	136	Single Number	LeetCode	-
☐	78	Subsets	LeetCode	-
☐	46	Permutations	LeetCode	-
☐	51	N-Queens	LeetCode	-

How to use it

- **Phase 1 (sections 1-6):** the first few weeks, daily. Aim to solve rating 800-1500 problems in under 30 minutes.
- **Phase 2 (sections 7-10):** DP is the part that fades fastest, so redo any problem you needed the editorial for after 3 days.
- **Phase 3 (sections 11-13):** graphs and trees. Write DSU, Dijkstra, BFS and a segment tree from scratch at least once without looking.
- **Rule of thumb:** if you're stuck for 40+ minutes, read the editorial hint (not the solution), then re-solve it a few days later.